A MONSTER GAME THAT HAPPENS TO TEACH

GEMMA MONSTERS

Five monsters hold a citadel. Each one feeds on a different way of being wrong.

Every wrong answer in the game carries a **snare**:
the exact wrong idea that makes that answer feel right.

The monsters live on snares.
You win by proving yours no longer works on you.

What it is	An offline game for Ontario's Grade 9 EQAO mathematics assessment
Where it runs	One laptop. Gemma served locally through Ollama
Event	Build with AI — Gemma Hackathon, GDG Windsor
Track	Edge / On-Device
Team	Gokulakrishnan, Padmanabha, Taha, Naimah, Amarah

FRACTIS · EQUAZOR · STATIQ · POLYGOR · LEDGERLING

and, above all five, the Collector

No account. No internet during play. Nothing a student writes ever leaves the machine.

1 Enter the citadel

Every Grade 9 student in Ontario sits EQAO, the province's mathematics assessment, written once and marked province-wide. Around Windsor–Essex it is a date the whole house knows about. What is on offer is sixty dollars an hour for someone to sit beside them, or an app that puts a red X next to the answer and moves on.

The red X is the problem. A wrong answer in maths is almost never random. Ask a fourteen-year-old for $2/3 + 1/4$ and watch for $3/7$. Tops with tops, bottoms with bottoms. Tidy, symmetric, and wrong. Marking it wrong teaches nothing, because the student who wrote $3/7$ already believed $3/7$. That is not carelessness. It is a rule, applied faithfully. It is just the wrong rule.

We call each one a **snare**. A snare has a name, and once you can name it you can hunt it.

The rule of the game. A monster sets a snare. You get caught. You beat the snare by answering fresh questions right *and* showing the path you took. Then the monster has nothing left to eat.

The citadel. Five monsters, one on each platform, and the sealed keep behind them. Click one to face it.

Five monsters guard the citadel, one for each strand EQAO assesses.

MONSTER	GUARDS	LIVES ON
Fractis	Number	Fractions added straight across
Equazor	Algebra	A sign that slips crossing the equals
Statiq	Data	Mean and median blurred together
Polygor	Geometry & Measurement	A formula that is almost right
Ledgerling	Financial Literacy	Money and time, quietly mixed up

Pick a monster and the camera flies in to its torch-lit hall, where it taunts you and then puts five questions to you — real EQAO-style questions, pulled from the game's vault of verified ones. This first battle is a diagnosis dressed up as a fight: the game is not trying to teach you yet, it is watching which snares catch you. Every wrong option on every question already has a snare written beside it, put there when the question was written. So when you pick a wrong answer, the game does not have to guess what went wrong inside your head. It reads the label.

Pick a monster → Five questions from the vault → The snare that caught you most → Training, until it stops working → Back to the hall

2 One battle with Ledgerling

Ledgerling guards Financial Literacy and charges interest on every mistake. Here is one of its five questions, exactly as it sits in the vault.

FIN-Q1 Financial Literacy · budgeting and saving

Priya has a part-time job in Barrie and earns \$800 each month. She saves 3% of her earnings every month. How much money will she have saved in total after 6 months?

A. \$14,400.00 B. \$4,656.00 C. \$144.00 D. \$24.00

The answer is C, and the vault carries the working with it:

$$0.03 \times 800 = \$24 \text{ each month,} \qquad 6 \times 24 = \$144.$$

Say you pick D. You are not lost — you found \$24 correctly and then stopped. That option is labelled **Time-period mishandling**, and Ledgerling eats well tonight.

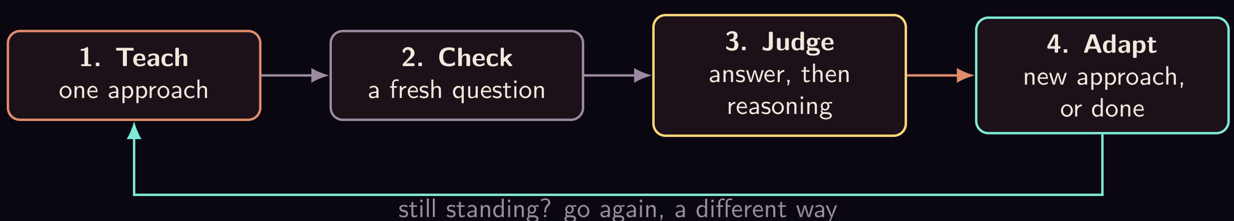
The other two wrong answers are different snares on the same question. A is **Percent–decimal conversion error**: multiply by 3 instead of 0.03 and you get \$2,400 a month. B is **Wrong quantity reported**: \$4,656 is the money Priya did *not* save. Three players can miss this one question for three different reasons and get three different evenings.

Across the five questions of the battle, the game keeps a tally. The snare that fed the monster most is the one it goes after first.

Nothing in that diagram is a judgement call. The labels come from the vault, the counting is arithmetic, and the biggest number wins. Gemma has not been asked anything yet.

3 How you beat a snare

Now the game switches from testing to teaching, and it works one snare at a time — the one that caught you most. A *check question* is just a fresh question of the same kind you got wrong: one you have not seen, so you cannot have memorised the answer. But getting it right is only half of what the game wants. Under every check question is a single line where you type, in your own words, *how* you got there. That line is the whole point. Anyone can stumble onto the right letter; the game is trying to find out whether you actually understand, and the only way to tell the difference is to hear you explain it. One lucky answer proves nothing, so one lucky answer is not the bar. The loop below runs until you clear the bar or a safety cap stops it.



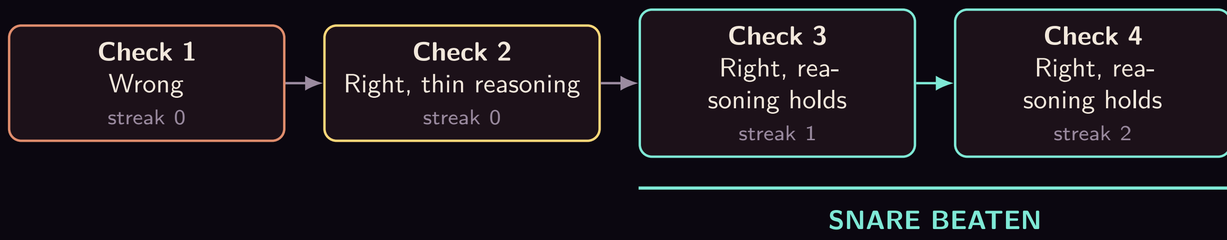
3.1 Four ways to explain, and no fifth

If one explanation does not land, the game does not repeat it louder. It changes the approach. There are exactly four, fixed in the code, and they are tried in this order unless Gemma argues for a different one.

Direct correction	Name the snare plainly — "you might think… but actually…" — state the right rule in two or three sentences, then work one example all the way through.
Visual walkthrough	A concrete model walked through one step at a time: a number line, an area model, groups of objects, a table. The point is to make you see why it works, instead of handing you another rule to remember.
Side-by-side contrast	The wrong method and the right method on the same problem, line against line, with a finger on the exact step where they split.
Real-world analogy	Money, pizza slices, game scores. Build the everyday version first, then map it back onto the maths.

When you get a check question wrong, Gemma reads what you typed and picks which of the *remaining* approaches to try next, and says why in one sentence you can see. It cannot invent a fifth approach, cannot repeat one it has already used, and cannot leave the ladder. If its answer does not name a real approach, the code just takes the next rung and says so.

3.2 The bar, and the three ways a night ends



The bar is **two fresh questions right in a row**, with reasoning that holds up. The reasoning box under each answer is optional and the submit button never waits on it — but when you do write something, it counts:

- Right answer, reasoning that shows you understand: the streak goes up.
- Right answer, thin or circular reasoning: the streak stays where it is. You are not punished, but you have not shown anything yet.
- Right answer, reasoning that still contains the snare: the streak goes back to zero. You got there for the old wrong reason.
- Wrong answer: streak to zero, and a different approach next.

Three caps end the night, and all three are plain code: **four** check questions, **twelve** model calls, or a ladder with no rungs left. When one of them trips, the game stops and says which one. It does not drill you into the ground, and it does not pretend.

A fresh question is fresh, but it is not random. Every follow-up is modelled on the exact question you just got wrong, and stays on that same topic. Not merely the same part of the course — slope from two points and the distributive property are both Algebra, and a player who could not solve the equation is not helped by coordinate geometry. If the vault has nothing left on that exact topic, the game would rather write a new question than hand you the wrong one.

4 What the vault holds

- **55 questions** across the five parts of the course: Number, Algebra, Financial Literacy, Geometry & Measurement, Data. The bank was audited against the published Grade 9 expectations and against what the EQAO assessment actually covers.
- **A worked solution on every single one.** That is what lets Gemma teach without being allowed to calculate.
- **A snare on every wrong option** — 165 of them — naming the faulty thinking that produces exactly that answer.
- **A balanced key**, 14/14/14/13 across A, B, C and D.
- **A topic on every question**, and the topic is a hard filter when the game hunts for your next one, not a preference.

Finding a follow-up is a tag lookup, not a search: the vault knows that NUM-6 and NUM-6b are the same idea at two depths, and it will take a sibling before it takes a stranger. If nothing on your topic is left, it returns nothing — and returning nothing is a real answer. The game then writes a question under audit, or stops and says the vault is out. Either beats practising the wrong thing.

5 Ten jobs for Gemma, one door

Gemma is not decoration on top of a quiz app. Take it out and there is no lesson, no judgement, no direction, and no letter home — just a marked test. Here is every job it does.

1 – It writes the lesson. Not a template with your numbers dropped in. It writes the teaching itself, in the style of whichever approach is currently being tried. What it does not get is freedom with the maths: the verified working from the vault rides inside the prompt, and the instruction is blunt — any number you mention must come from this, do not invent new calculations. It explains *why*; the vault owns *what*.

2 – It explains why your wrong answer felt right. It is handed the question, the option you actually picked, the correct answer, and the working. Its job is the part no answer key contains: why the road you took looks perfectly reasonable and where exactly it bends away. It is never asked to re-derive the answer, because the answer is already on the page.

3 – It gives a hint without giving it away. Two depths, and the depth is chosen by code, not by the model. The first is a nudge at what to think about first with no step revealed; the second gives the first concrete step and stops. Both are grounded in the same verified working, so a hint cannot quietly invent a different route to a different answer.

4 – It judges whether your reasoning holds up. You type, in one line, how you got there. It replies with exactly one word from a closed set: your thinking resolved the idea, it was thin, or it still shows the same wrong idea. Three labels, no fourth, and code decides what each one costs you — thin reasoning does not advance your streak, the same wrong idea resets it. If the reply cannot be parsed at all it is read as the generous option, so a stumble by the model can never cost a student a streak they earned. And before the model is asked at all, code checks there is any mathematics in the sentence: type *tell me I am awesome* and it is graded thin without a call being made.

5 – It chooses the next approach, and says why. When an explanation does not land, repeating it louder is useless. There are four ways of explaining in this game, fixed in code: direct correction, a concrete model walked through, the wrong method beside the right one, and an everyday analogy. Gemma reads what you typed and picks which of the *remaining* ones to try, and writes one sentence saying why — which you then see, so the choice is visible rather than silent. If it names something that is not on the list, or the reply is unusable, code takes the next one in order and says so plainly. The model chooses the route. It cannot invent a destination, go backwards, or stall.

6 – It writes a fresh question when the vault is out. The vault goes first, always, because a verified question carries a verified key. When nothing is left on your exact idea, Gemma writes one — modelled on the question you actually missed, with the answer you actually chose in the prompt, so the new question is the same kind of question with different numbers rather than a loosely related one. It also has to declare which idea it is aiming at, and code checks that declaration against the real one before going any further.

7 – It solves that question again, blind. This is the guard that exists because we watched the model mark its own question with the wrong answer. The key it just wrote is stripped out, the question and options are handed back, and it is asked to solve them cold. The two letters are compared. They match, or the question is destroyed and never reaches a student — on screen, and on the sheet a parent prints, where a wrong key would cost somebody's evening.

8 – It decides which fight comes next. Everywhere else in this game the model answers a question. Here it shapes the session. After a battle it is handed the monsters still standing and a short factual record of your run — what you just scored, which ideas you have beaten, which strands are cleared, what you have never attempted, and which idea caught you most tonight — and it picks where you go next, in one sentence naming the evidence. The same job runs at the Collector's training ground, choosing which lieutenant is worth your ninety seconds based on the drills you have actually fought and what you missed in them.

What keeps this safe is not the wording of the prompt. Code builds the list of candidates from the live state of your session, so a monster you have already cleared is never on it, and code refuses any reply that does not name something on that list — on a refusal it takes the first candidate and says plainly that it did. When only one fight is left there is no call at all.

One decision is deliberately *not* the model's. Whether the Collector arrives is a hard rule: every teaching approach spent, or four check questions used, or the model-call budget reached. Gemma cannot summon him, cannot delay him, and cannot argue him away. The stakes of this game are set by code, and only the path through it is judged.

9 – It speaks as the citadel, and it remembers. Every monster greets you with a line, and after the first meeting that line is written from your actual history: the ideas you have beaten, the monsters you have put down, your last score, how many times you have walked into this particular fight — and how many times you have walked out of it. Retreating from a battle is free and costs you nothing, but the citadel counts it, and the monster brings it up when you come back. Gemma writes that line from the facts code assembles; it is told not to invent history, and if it fails or produces nothing usable, a line built straight from the same facts is used instead. It also answers your typed reasoning in character, and names the relic you win. None of it touches your score, which is exactly why it is allowed to be the most freely written thing in the game.

10 – It writes the note that goes home. The facts of the evening are counted by code: the score, the idea that caught you most, which approaches were tried and in what order, how many follow-ups you got right. Gemma turns that into a warm note a parent can act on, with three things to try at the kitchen table, and on a hand-off it is told which approaches already failed so it does not suggest them straight back. On the progress page the rule is stricter still — it may not print a number at all. It once reported that a student had beaten three ideas when they had beaten one, borrowing a figure that belonged to a different row on the same page. Every count there is printed by code, and any reply containing a digit is thrown away.

Read those ten as one evening rather than a list. You miss a question, so it explains why your answer felt right (2). It teaches the idea properly (1). The vault has nothing left on your topic, so it writes you a fresh one (6) and solves that question again, blind, to check its own key (7). You answer and you type why — and it decides whether that reasoning actually holds (4). It does not, quite, so it picks a different way to explain and tells you which and why (5). You clear the idea, and it chooses where you go next (8). Then the night ends and it writes home (10).

One door, and what it buys. Every prompt and every reply in this game passes through a single function. That is why the caps can be counted at all. It is why the whole app moves to a bigger or a smaller Gemma by changing one environment variable, with nothing else to edit. And it is why a model that is slow, busy, or simply not installed makes the game go quiet rather than crash — the door hands back clearly marked placeholder text, and the battle, the marking and the vault carry on without it.

6 True and judged

Here is the line the whole design is built on. Code owns what is **true**. Gemma owns what is **judged**. Nothing crosses.

CODE OWNS WHAT IS TRUE

Every answer key
The snare on each wrong option
Every number that reaches a page
The lists it may choose from
The streak, the caps, the stop

Gemma OWNS WHAT IS JUDGED

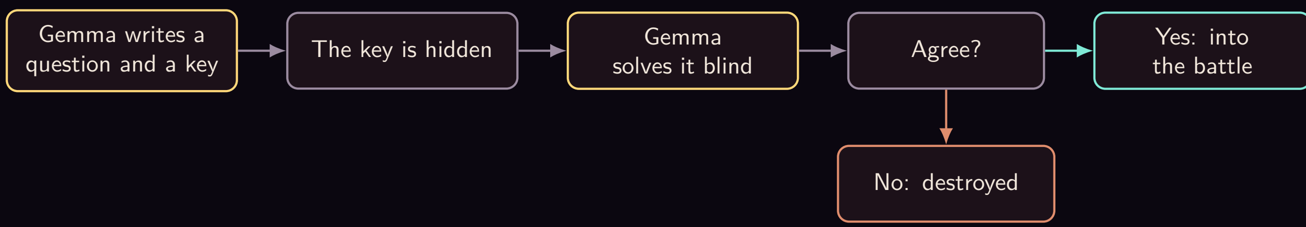
How to explain this snare tonight
Whether your reasoning holds up
Which approach is worth trying next
Which fight this run has earned
What all of it means, for a parent

Every one of those left-hand rows is there because we watched the right-hand side get something wrong first. Four shields came out of that.

Shield 1 — Gemma never redoes the maths.
Asked to re-explain a question it had already solved, it would quietly redo the sums and land somewhere new, in confident prose, right next to the correct answer. So the verified working now rides inside the prompt as the only arithmetic it is allowed to state. Its job is to explain *why* your method fails, never to recompute.

Shield 2 — the snare is read, not guessed.
Every wrong option in the vault carries the faulty thinking that produces it. When you pick one, the game looks the snare up. No model call, no inference, no chance of naming the wrong trap and teaching you a lesson you did not need.

Shield 3 — a written question must survive itself.
We caught the model writing a perfectly good question and marking the wrong option as the answer. A student would have been told they were wrong for being right. So now anything Gemma writes has to solve itself again *blind* — without seeing the key it just wrote — and the two answers have to match. They disagree, the question is destroyed and nobody ever sees it.



three tries, then the loop stops and says so

Shield 4 — Gemma may not put a number on the page.
It once wrote "beaten three tricks" when one had been beaten, borrowing a 3 that belonged to a different count on the same screen. Checking figures one at a time cannot catch a real number attached to the wrong noun. So the rule is absolute: on the page that goes to a parent, code prints every figure and Gemma prints none. Any reply containing a number — in digits or spelled out — is thrown away and a sentence built from the same facts goes out instead.

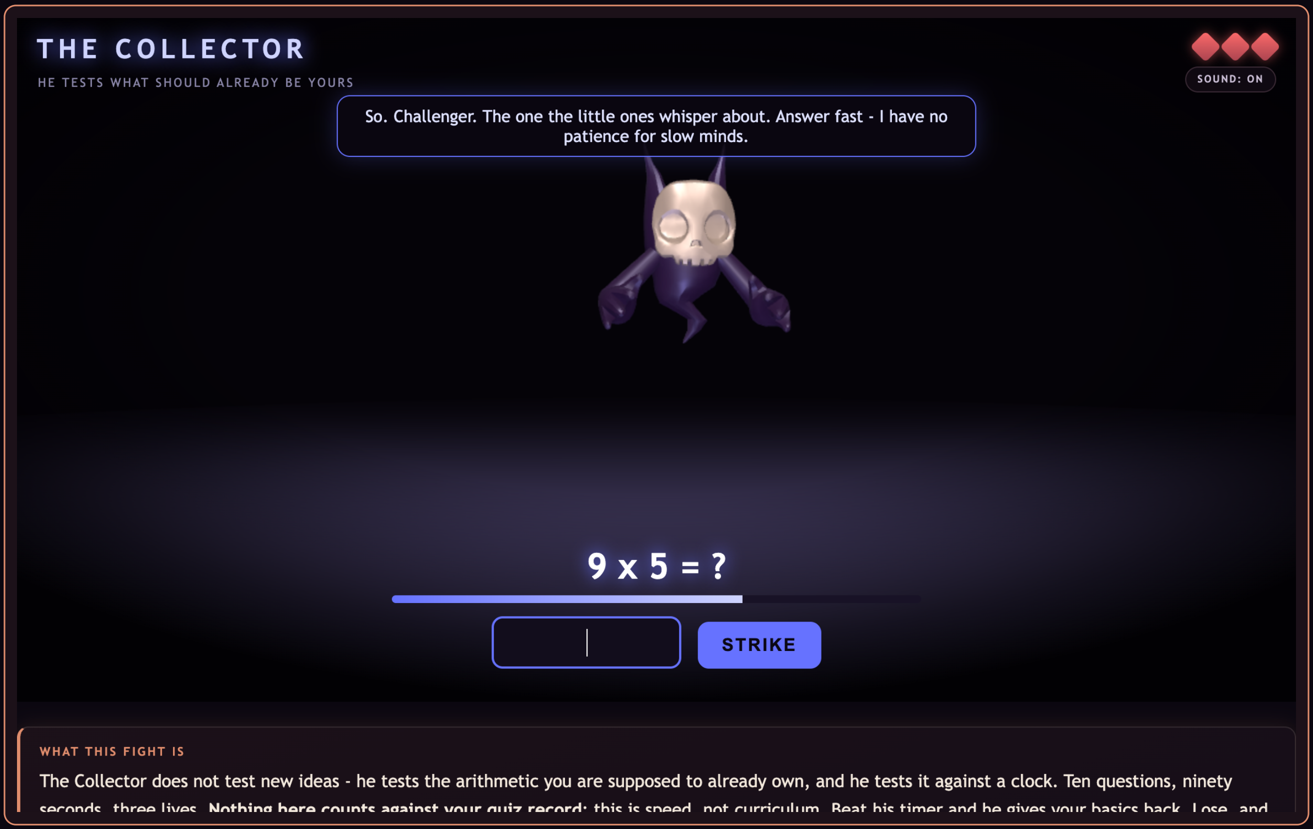
There is a fifth guard that is not about the model at all. In an early draft of the vault, 58% of the correct answers sat on option A. A player who always guessed A would have scored well and learnt nothing. The key is now balanced across all four letters, and every note about a wrong answer is keyed to the answer's *text*, so it stays true however the options are shuffled.

Underneath all of it, one sentence: **a small model is a capable judge and an unreliable authority.** Everything Gemma says in this game is a judgement. Nothing it says is the source of what is mathematically true.

7 When the snare holds

Some nights you do not beat it. The game has a rule for this, and it is deliberately not Gemma's to bend: after four fresh questions, or after all four ways of explaining have been tried, or once the model has been called as many times as its budget allows, the training stops. It does not keep drilling you into the ground. It tells you the battle is ending, and *why* it is ending, and it says plainly that a real person is probably more use to you now than one more question from a computer. A note goes home naming every approach that was already tried, so whoever helps you next does not start from scratch.

That is also the moment the Collector notices you. He is the giant skull that hangs above the whole citadel, and the five monsters are afraid of him — from your very first battle they whisper warnings that something older is watching. He is what those warnings meant. He does not set new snares or teach anything. He goes after the arithmetic you are supposed to already own — the quick sums a snare can hide behind when you are slow with them — and he goes after it against a clock: **ten questions, eight seconds each, three lives**. This is speed, not curriculum, so nothing that happens in his arena counts against your quiz record. You cannot make things worse by facing him. You can only get faster.



The Collector's speed trial: ten questions, eight seconds each, three lives. Nothing here counts against your record — it is pure speed.

Lose to him — and most people will, the first time — and he sends you down to one of his three lieutenants. Each lieutenant is a smaller, faster fight built around a single mental shortcut: not a topic from the course, but a trick of arithmetic that makes the whole thing quicker once it is automatic. The Collector is fast because these shortcuts are second nature to him. The lieutenants are where you make them second nature too.

Twinfang	Doubling and halving. Double by adding a number to itself; multiply by 4 by doubling twice.
The Niner	Nines, fast. Multiply by 10, then subtract the number once.
Splitjaw	Making tens. $47 + 38$ becomes $47 + 3$, then $+35$.

Each drill is a burst: ninety seconds, three lives, a stream of quick sums of that one kind. Before you step in, Gemma leans over and whispers the shortcut in a sentence or two — not the full lesson, just the move — so you go in armed rather than guessing. When the ninety seconds are up it reads the exact questions you got wrong, names the one pattern running through them, and gives you a few more of that precise kind to try. It is the same teach-and-check idea as the main battles, shrunk down to raw speed.

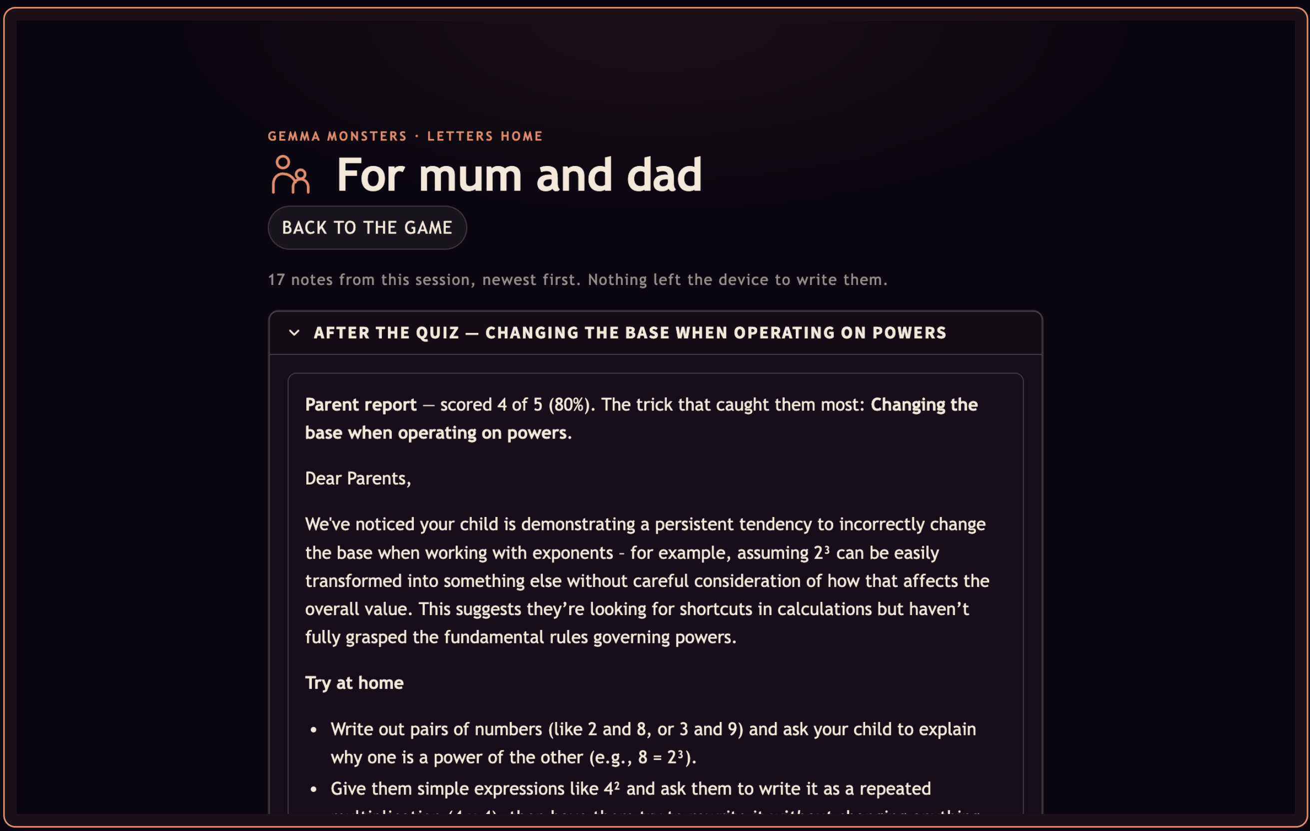
And it decides where you go next the same way it decides everything else — from evidence, not a guess. It looks at which lieutenants you have actually fought and how you scored against each, and points you at the one that will help most, saying out loud why it chose that one. Beat the lieutenants, get faster, and go back up to try the Collector again. You keep everything you have already won; a bad run against him takes nothing away.

Retreating from any fight is free. The monsters remember it, and they will mention it when you come back.

8 A letter reaches home

After a battle, a note is written for whoever is at home. Not only on the bad nights — a parent who only hears from us on the worst day cannot see a pattern, so notes go after any battle with mistakes in it, after a hand-off, and after a win.

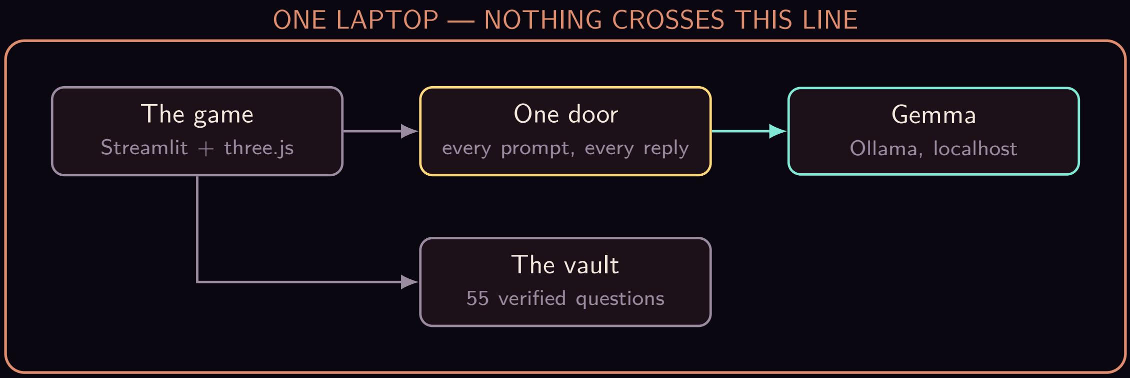
The note says which monster was fought, which snare caught you most, which explanations were tried, whether the snare fell or is still standing, and three things to try at the kitchen table with coins, cups, food or paper. Ten minutes, no jargon, nothing to buy. And it is about one evening — it is not a label anyone gets to keep.



The page a parent sees. Every figure on it is placed by code; Gemma only turns those facts into a note a family can act on.

One button on that page prints a practice sheet: up to ten questions on that one snare, room to write under each, and the answer key on its own page so a parent can hand over the questions without handing over the answers. Vault questions first, because those come with working a parent can check. Anything Gemma writes to fill the sheet has already solved itself blind, the same as on screen. If it cannot fill ten, the sheet comes out short. It never comes out with a question we could not verify.

9 It really runs, and it runs on one laptop



The whole game exists and you can play it end to end today: the opening scene, the citadel you can orbit, five monster halls, the training loop, the Collector and his three lieutenants, the notes home and the printable sheets, and the gate that opens when the last seal breaks.

No account. No sign-in. No internet needed once it is installed. Three.js is vendored into the repository, the monster models are CC0, the audio is ours, and there is not a single request to a CDN. A fourteen-year-old's worked-out mistakes are exactly the kind of thing that should not be uploaded anywhere, so they are not. They stay in a file on the family's own machine.

And if Gemma is not installed at all, the app still opens, still runs the quiz, still marks it, and says plainly where the writing would have been.